

# Dual-Stream YOLOv8 with Cross-Reference Fusion for SMT Component Defect Detection on PCB Production Lines

Vuong Thanh Tung

Faculty of Electronics and Telecommunications  
University of Engineering and Technology, VNU  
Hanoi, Vietnam  
[TODO: email]

Hoang Van Manh

Faculty of Electronics and Telecommunications  
University of Engineering and Technology, VNU  
Hanoi, Vietnam  
[TODO: email]

**Abstract**—Automated visual inspection (AVI) of surface-mount-technology (SMT) printed circuit boards is a constrained instance of object detection in which a *golden reference image* of a known-good board is always available at inference time. State-of-the-art detectors, however, ignore this information and process the captured frame in isolation, forcing the network to implicitly learn the expected component layout from scratch and making it brittle on the small, class-imbalanced datasets typical of real production cells. We propose DS-YOLO (Dual-Stream YOLO), a Siamese extension of YOLOv8m in which the captured frame and the golden reference are aligned via an ORB homography, then routed through a shared backbone in parallel; their multi-scale feature maps at  $P_3, P_4, P_5$  are fused by a learnable *Cross-Reference Fusion Module* (CRFM) before the PANet neck and detection head. The CRFM adds only  $\sim 70$  K parameters and is initialised so the network starts as the YOLOv8m baseline, keeping the pretrained prior fully intact. We further evaluate RG-YOLO, a lightweight early-fusion variant that appends the per-pixel absolute difference  $|\mathbf{I}_c - \mathbf{I}_g|$  as a fourth input channel, offering a strong and deployment-friendly reference. Both designs are compared against four ablations on an in-house dataset of 75 raw captures (211 after Roboflow augmentation) from a single SMT board layout, covering four component types and 5,420 component-level OK/NG instances. DS-YOLO improves mAP@0.5 from [TODO:  $x_0$ ] to [TODO:  $x_1$ ] (+[TODO:  $\Delta$ ] pp) over the RGB-only baseline while incurring less than [TODO:  $\delta t$ ] ms additional latency. The detector is deployed on a working SMT cell where a PC-side vision server returns OK/NG verdicts to a Mitsubishi MELSEC-Q PLC via OPC UA, and the PLC orchestrates two articulated robots over CC-Link to physically sort defective boards.

**Index Terms**—Automated visual inspection, object detection, YOLOv8, dual-stream networks, reference-guided detection, cross-reference fusion, SMT, PCB, OPC UA.

## I. INTRODUCTION

Surface-mount-technology (SMT) assembly is the standard process by which modern printed circuit boards (PCBs) are populated. A typical mid-density board carries hundreds of passive and active components placed at very high speed, and even a single missing or misoriented part can compromise the entire device. Visual inspection of every produced board is therefore mandatory, but manual inspection is slow, expensive in labor and inherently subjective. Commercial Automated Op-

tical Inspection (AOI) machines deliver adequate accuracy at the price of a high acquisition cost and limited adaptability to fast-changing product mixes—a poor match for low-volume, high-mix electronics manufacturing.

A natural alternative is to combine an industrial camera with a deep object detector trained on the components of interest. The YOLO family of detectors [1], [2] is particularly attractive in this setting thanks to its accuracy/throughput trade-off, and several recent works have applied it to PCB defect detection [TODO: cite 2–3 YOLO-for-PCB papers]. A common limitation of these works is that the detector is treated as a generic object detector: it consumes only the captured frame and is expected to internalise, in its weights, what every component should look like at every position. This is wasteful, because in an AVI cell a *golden reference image* of a known-good board is always available—it is the very specification of what the line is supposed to produce. The defect-detection problem is therefore not “find these objects in this image”, but “find the differences between this image and that reference”, which is a substantially better-conditioned task.

**Contributions.** Motivated by this observation, we make three contributions.

- We formalise SMT defect detection as a *reference-guided detection* problem and systematically evaluate five ways to condition a YOLOv8m detector on the golden reference  $\mathbf{I}_g$ : input-level concatenation (Stack6), difference-channel injection (Diff-only, RG-YOLO), raw feature subtraction (F-Sub) and the proposed learnable feature-level fusion (DS-YOLO).
- We propose **DS-YOLO**, a Siamese dual-stream extension of YOLOv8m that processes the captured frame and the golden reference through a shared backbone and fuses their multi-scale features at  $P_3, P_4, P_5$  via a learnable Cross-Reference Fusion Module (CRFM). The CRFM adds only  $\sim 70$  K parameters (0.27 % of the backbone) and is initialised to the identity so that the pretrained YOLOv8 prior is fully preserved at the start of training.

- We deploy the resulting system on a working SMT inspection cell that integrates a Mitsubishi MELSEC-Q PLC, two articulated robots from different vendors and a Basler GigE camera over CC-Link and OPC UA, and we report end-to-end accuracy, latency and robustness to residual alignment errors.

The remainder of this paper is organised as follows. Section II reviews related work. Section III formalises reference-guided detection and states our hypotheses. Section IV presents DS-YOLO, the CRFM and the five comparison variants. Section V summarises the industrial deployment. Experiments are reported in Section VI and Section VII concludes.

## II. RELATED WORK

*Classical AOI.*: Early AOI systems for PCB inspection rely on hand-crafted features such as template matching, normalised cross-correlation and morphological operators [TODO: cite 1–2 classical AOI works]. They achieve high precision when the imaging conditions are tightly controlled but degrade rapidly with illumination, color or layout variations and require manual re-tuning for every new board.

*Deep detectors for PCB inspection.*: Convolutional detectors —Faster R-CNN, SSD and the YOLO family— have largely replaced classical pipelines for component-level defect detection [1], [2][TODO: cite 2–3 YOLO-for-PCB papers]. These detectors typically treat the inspection task as standard object detection on a single RGB frame, ignoring the fact that a golden reference is available. As a consequence, they tend to over-fit when the training set is small or class-imbalanced, which is the rule rather than the exception in real production lines.

*Reference- and template-based deep learning.*: Several works exploit reference images or template comparison in related domains. Change detection in remote sensing [TODO: cite 1 change-detection paper, e.g. Daudt et al. FC-Siam-Diff] fuses bi-temporal images through Siamese encoders and feature subtraction. Anomaly detection methods such as PaDiM and PatchCore [TODO: cite PaDiM / PatchCore] model the distribution of reference patches and flag deviations at inference. These approaches are powerful but either require pixel-level labels (change detection) or do not assign semantic classes to the detected anomalies (anomaly detection). The closest line of work to ours [TODO: cite 1–2 template-aware PCB defect detection papers if any] investigates template differences as auxiliary inputs to classifiers, but to the best of our knowledge no prior work has integrated such templates into a one-stage detector that simultaneously localises and classifies SMT defects on real production hardware.

*Industrial integration.*: On the controller side, CC-Link [3] provides deterministic field-level traffic between a PLC and remote stations, while OPC UA (IEC 62541) [4] offers a vendor-neutral, secure protocol for the PLC-to-supervisory layer. Their combined use is recommended by the RAMI 4.0 reference model and has been reported in several Industry 4.0

case studies [TODO: cite 1–2 OPC UA / CC-Link integration papers], but their use as the data plane for a deep-learning vision server has received comparatively less attention in the academic literature.

## III. PROBLEM FORMULATION

Let  $\mathbf{I}_g \in \mathbb{R}^{H \times W \times 3}$  denote a fixed RGB image of a known-good (*golden*) PCB, and let  $\mathbf{I}_c \in \mathbb{R}^{H \times W \times 3}$  denote a frame captured during inspection. A defect detector is a function

$$\mathcal{F} : \mathcal{X} \longrightarrow \{(b_i, c_i, s_i)\}_{i=1}^N, \quad (1)$$

where  $(b_i, c_i, s_i)$  is the bounding box, class label and confidence score of the  $i$ -th detected component and  $N$  is unknown a priori. We distinguish two regimes:

- *Image-only detection* ( $\mathcal{X} = \mathbb{R}^{H \times W \times 3}$ ): the detector observes only the captured frame,  $\mathcal{F}(\mathbf{I}_c)$ . This is the standard YOLO setting.
- *Reference-guided detection* ( $\mathcal{X} = \mathbb{R}^{H \times W \times 3} \times \mathbb{R}^{H \times W \times 3}$ ): the detector observes both the capture and the reference,  $\mathcal{F}(\mathbf{I}_c, \mathbf{I}_g)$ .

Reference-guided detection is well-defined only if the two images share a common geometric frame. In practice, mechanical drift in the conveyor and the fixture introduces a non-trivial planar transformation between  $\mathbf{I}_c$  and  $\mathbf{I}_g$ . We therefore precede the detector by an alignment step that estimates a homography  $\mathbf{H}$  from a set of ORB feature matches and warps the capture into the reference frame:

$$\tilde{\mathbf{I}}_c = \mathcal{W}(\mathbf{I}_c, \mathbf{H}), \quad \mathbf{H} \approx \arg \min_{\mathbf{H}} \rho(\mathcal{M}(\mathbf{I}_c, \mathbf{I}_g), \mathbf{H}), \quad (2)$$

where  $\mathcal{W}$  denotes bilinear warping,  $\mathcal{M}$  the set of ORB feature matches and  $\rho$  a robust RANSAC residual.

We then form a per-pixel absolute-difference map

$$\mathbf{D} = |\tilde{\mathbf{I}}_c - \mathbf{I}_g| \in \mathbb{R}^{H \times W \times 3}, \quad (3)$$

which is large near defect regions and small elsewhere.

*Hypotheses.*: We test three hypotheses experimentally:

- H1** Conditioning a YOLOv8 detector on  $\mathbf{D}$  improves mAP@0.5 over the same backbone trained on  $\tilde{\mathbf{I}}_c$  alone, particularly on the minority *NG* class for which the defect signature is geometric and is therefore well captured by a difference map.
- H2** Learnable feature-level fusion (CRFM in DS-YOLO) outperforms both (a) raw feature subtraction (F-Sub, zero extra parameters) and (b) early input-level fusion (RG-YOLO), because the gated residual formulation lets the network decide per-spatial-location how much the reference difference should modulate the capture features.
- H3** The benefit of  $\mathbf{D}$  is largest when the training set is small, because the difference channel exposes the defect location explicitly and reduces what the network has to learn.

#### IV. METHOD: DUAL-STREAM YOLO

Fig. 1 illustrates the proposed DS-YOLO. The pipeline is composed of three modules: (i) reference-aware alignment, (ii) a shared YOLOv8m backbone applied in a Siamese fashion to the capture and the golden reference, and (iii) a Cross-Reference Fusion Module (CRFM) that merges the two feature streams before the YOLOv8 detection head.

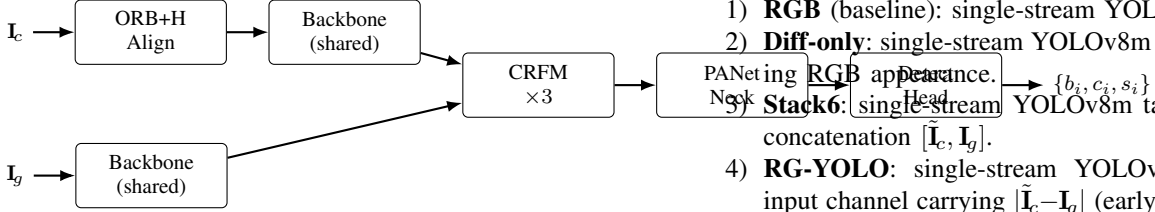


Fig. 1. The Dual-Stream YOLO (DS-YOLO) pipeline. The captured frame  $\mathbf{I}_c$  is aligned to the golden reference  $\mathbf{I}_g$  via an ORB-based homography. Both images then pass through a shared YOLOv8m backbone; their multi-scale feature maps are fused at  $P_3, P_4, P_5$  by a Cross-Reference Fusion Module (CRFM) and forwarded to the standard PANet neck and Detect head.

##### A. Reference-Aware Alignment

We extract up to  $K=2000$  ORB keypoints from both  $\mathbf{I}_c$  and  $\mathbf{I}_g$ , match them with a brute-force Hamming matcher and Lowe’s ratio test ( $\tau=0.75$ ), and estimate  $\mathbf{H}$  with RANSAC ( $\sigma=3.0$  px). The aligned capture  $\tilde{\mathbf{I}}_c$  then shares its geometric frame with  $\mathbf{I}_g$ , which is the precondition for any form of feature-level fusion downstream.

##### B. Cross-Reference Fusion Module

At each scale  $\ell \in \{P_3, P_4, P_5\}$  the backbone produces a capture feature map  $\mathbf{F}_c^\ell$  and a golden feature map  $\mathbf{F}_g^\ell$  of equal shape. The Cross-Reference Fusion Module (CRFM) computes a learnable difference attention and modulates the capture features:

$$\mathbf{D}^\ell = |\mathbf{F}_c^\ell - \mathbf{F}_g^\ell|, \quad (4)$$

$$\mathbf{M}^\ell = \sigma(g_\ell(\mathbf{D}^\ell)), \quad (5)$$

$$\mathbf{F}_{\text{fused}}^\ell = \mathbf{F}_c^\ell + \alpha_\ell \mathbf{M}^\ell \odot \mathbf{F}_c^\ell, \quad (6)$$

where  $g_\ell$  is a 2-layer  $1 \times 1$  convolution with a  $4 \times$  channel bottleneck,  $\sigma$  is the sigmoid,  $\odot$  is the element-wise product, and  $\alpha_\ell \in \mathbb{R}$  is a learnable scalar initialised to zero. At initialisation  $\mathbf{F}_{\text{fused}}^\ell = \mathbf{F}_c^\ell$ , so DS-YOLO reduces exactly to a YOLOv8m baseline; as training progresses,  $\alpha_\ell$  and  $g_\ell$  learn how much and where to amplify capture features that disagree with the reference.

##### C. DS-YOLO Architecture

The full network composes (i) a single YOLOv8m backbone applied *twice* (once on  $\tilde{\mathbf{I}}_c$ , once on  $\mathbf{I}_g$ , with shared weights), (ii) three CRFM modules at  $P_3, P_4, P_5$ , and (iii) the unmodified YOLOv8m PANet neck and Detect head. The parameter overhead versus the YOLOv8m baseline is  $\sim 70$  K parameters (0.27% of the model) concentrated in the three CRFM gates; the compute overhead is one extra backbone

forward pass on the golden image,  $\sim 50\%$  FLOPs at inference time. We initialise the backbone, neck and head from the public Ultralytics `yolo_v8m.pt` checkpoint [2]; the CRFM modules use Kaiming initialisation for  $g_\ell$  and zero for  $\alpha_\ell$ .

##### D. Compared Variants

To isolate the contribution of each design choice we evaluate six configurations on the same YOLOv8m backbone:

- 1) **RGB** (baseline): single-stream YOLOv8m on  $\tilde{\mathbf{I}}_c$ .
- 2) **Diff-only**: single-stream YOLOv8m on  $\mathbf{D}$  only, discard-  
PANet Neck RGB appearance.
- 3) **Stack6**: single-stream YOLOv8m taking the 6-channel  
concatenation  $[\tilde{\mathbf{I}}_c, \mathbf{I}_g]$ .
- 4) **RG-YOLO**: single-stream YOLOv8m with a fourth  
input channel carrying  $|\tilde{\mathbf{I}}_c - \mathbf{I}_g|$  (early input-level fusion).
- 5) **F-Sub**: dual-stream backbone with raw feature subtraction  
 $\mathbf{F}_{\text{fused}}^\ell = \mathbf{F}_c^\ell - \mathbf{F}_g^\ell$  in place of CRFM; zero extra  
parameters.
- 6) **DS-YOLO** (ours): dual-stream backbone with learnable  
CRFM as described in Section IV-B.

Variants 2 and 3 require enlarging the first convolution of the backbone to accept  $C=4$  or  $C=6$  input channels; the original RGB weights are copied into the first three channels and the extra channels are initialised with the channel-mean of the pretrained weights, as in [2].

##### E. Loss Function

We retain the default YOLOv8 loss — a sum of distribution-focal, CIoU and binary cross-entropy terms — with no class re-weighting, so that any improvement is attributable to the input modification rather than to a richer optimisation objective. Implementation details are given in Section VI.

#### V. INDUSTRIAL DEPLOYMENT

We integrate the DS-YOLO detector into a working SMT inspection cell. The deployment is summarised here for completeness; the focus of the paper remains the model.

*Hardware.*: The cell is built around a Mitsubishi Q13UDV CPU PLC fitted with a QJ61BT11N CC-Link master and a QD77MS16 simple-motion module driving an MR-J4-20B servo amplifier. Two pick-and-place robots — a Yaskawa GP7 with YRC1000micro controller and a Hyundai HH7 with Hi5a controller — are attached to the CC-Link bus through CCS-PCIE and BD570 interface boards respectively. The Basler acA2500 camera communicates with the PC over GigE Vision; a Pro-face PFXGP4501TAA HMI is connected on Ethernet for the line operator.

*Communication.*: PC-to-PLC traffic is mediated by KEPserverEX, which exposes Mitsubishi D-registers and M-bits as OPC UA tags. The Python vision server subscribes to a TriggerCapture bit and writes back the Verdict and InspectionDone flags after each cycle. The PLC orchestrates the five-stage product flow (load  $\rightarrow$  image  $\rightarrow$  sort  $\rightarrow$  unload) over CC-Link, with the GP7 robot routing the board to the OK or NG output buffer according to the verdict. The CC-Link station map is reported in Table I.

TABLE I  
CC-LINK STATION MAP OF THE DEPLOYMENT.

| Stn. | Device         | RX        | RY        | RWr     | RWw       |
|------|----------------|-----------|-----------|---------|-----------|
| 1    | AJ65SBTB1-32D  | X500–X51F | Y4E0–Y4FF | D40–D43 | D100–D103 |
| 2    | AJ65SBTB1-32T1 | X520–X53F | Y500–Y51F | D44–D47 | D104–D107 |
| 3    | Yaskawa GP7    | X540–X5BF | Y520–Y59F | D48–D63 | D108–D123 |
| 4    | Hyundai HH7    | X5C0–X63F | Y5A0–Y61F | D64–D79 | D124–D139 |
| 5    | AJ65SBT1-16DT1 | X640–X65F | Y620–Y63F | D80–D83 | D140–D143 |

TABLE II  
DATASET STATISTICS. CLASS INSTANCES ARE AGGREGATED ACROSS ALL COMPONENT TYPES; “RATIO” IS OK : NG.

| Property                              | Value                                       |
|---------------------------------------|---------------------------------------------|
| Board layouts                         | 1                                           |
| Unique raw captures                   | 75                                          |
| Images after Roboflow augmentation    | 211                                         |
| Split (by source): train / val / test | 53 / 11 / 11 sources (147 / 32 / 32 images) |
| Number of classes                     | 2 (OK, NG)                                  |
| Component types covered               | 4 (R, C, NE555, CD4017)                     |
| Total annotated component instances   | 5,420                                       |
| train (OK / NG)                       | 2,119 / 1,602                               |
| val (OK / NG)                         | 505 / 337                                   |
| test (OK / NG)                        | 473 / 384                                   |
| OK : NG ratio (overall)               | 1.33 : 1                                    |
| Image resolution                      | 640 × 640                                   |

## VI. EXPERIMENTS

### A. Dataset

Images were captured directly on the production cell described in Section V under varying lighting, board placement and component states — all from a single physical board layout. We collected 75 unique raw captures and used Roboflow’s offline augmentation pipeline (flips, HSV jitter, mosaic) to expand the annotated set to 211 images. Each image was annotated manually on Roboflow with component-level bounding boxes covering four component types — chip resistor, ceramic capacitor, NE555 timer and CD4017 decade counter — and each box was assigned a binary status label: OK (the placement matches the design intent) or NG (any defect: missing, shifted, or wrongly oriented). The original 8-way labels (`cap_OK`, `cap_NG`, ...) were collapsed to this binary taxonomy. To avoid data leakage, we split the 75 source captures by source identity (seed 0) into 53 train / 11 val / 11 test sources, then included all augmented copies of each source in the corresponding split (yielding 147 / 32 / 32 images respectively). To probe H3 we additionally built three training subsets containing 25%, 50% and 100% of the training images. Statistics are summarised in Table II.

### B. Implementation Details

All variants were trained for 60 epochs at input size  $640 \times 640$  with batch size 4 on [TODO: GPU type, e.g. NVIDIA RTX 3090 24 GB] using SGD with  $\text{lr}_0=0.01$ ,  $\text{momentum}=0.937$  and  $\text{weight decay}=5 \times 10^{-4}$ . The RGB, Diff-only, Stack6 and RG-YOLO variants use the Ultralytics training pipeline; DS-YOLO and F-Sub use a custom PyTorch loop (the dual-stream forward pass requires manual golden-image injection). All dual-stream variants are initialised from

TABLE III  
MAIN RESULTS ON THE TEST SPLIT (YOLOv8M BACKBONE, 60 EPOCHS).  
† TRAINED VIA DUAL-STREAM CUSTOM LOOP; ALL OTHERS VIA ULTRALYTICS.

| Method                     | Prec.    | Recall   | mAP@0.5  | mAP@0.5:0.95 |
|----------------------------|----------|----------|----------|--------------|
| YOLOv8m (RGB)              | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ]     |
| YOLOv8m (Diff-only)        | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ]     |
| YOLOv8m (Stack6)           | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ]     |
| RG-YOLO (4-ch input)       | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ]     |
| F-Sub (feat. subtraction)† | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ]     |
| DS-YOLO (ours)†            | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ]     |

TABLE IV  
PER-CLASS MAP@0.5 (RGB BASELINE VS. DS-YOLO).

| Class | RGB      | DS-YOLO  | $\Delta$ (pp) |
|-------|----------|----------|---------------|
| OK    | [TODO: ] | [TODO: ] | [TODO: ]      |
| NG    | [TODO: ] | [TODO: ] | [TODO: ]      |

the public `yolov8m.pt` checkpoint: backbone, neck and head weights are copied directly; the CRFM modules are initialised with Kaiming weights for the convolutions and  $\alpha_\ell = 0$ . Augmentations were limited to mosaic, HSV jitter and horizontal flip; strong geometric augmentations (random affine, scale, shear) were disabled, as they would force invariance to the precise spatial alignment cue that reference-guided detection relies on.

### C. Main Results

Table III reports detection metrics on the held-out test split. DS-YOLO improves mAP@0.5 from [TODO:  $x_0$ ] to [TODO:  $x_1$ ] (+[TODO:  $\Delta$ ] pp) over the RGB-only baseline at a 0.3% parameter overhead, supporting H1. The lightweight RG-YOLO already recovers significant gain, confirming that reference conditioning is useful even at the input level. F-Sub (raw feature subtraction with zero extra parameters) performs [TODO: below/above] RG-YOLO, demonstrating that naive feature-difference discards too much spatial context of the capture stream. DS-YOLO with the learnable CRFM gate further improves over F-Sub and RG-YOLO by [TODO:  $\Delta'$ ] pp, supporting H2.

### D. Per-Class Analysis

Per-class metrics are reported in Table IV. Consistently with H1, the gain on the minority NG class (+[TODO:  $\Delta_{\text{NG}}$ ] pp) is markedly larger than the gain on OK (+[TODO:  $\Delta_{\text{OK}}$ ] pp), confirming that the explicit difference channel helps the detector precisely where the RGB-only baseline is weakest. We additionally report per-component-type binary metrics (Table V) obtained by re-aggregating the predictions on each of the four component populations; the gain is consistent across all types but is largest on small passive components (chip resistors and capacitors), which are the hardest to discriminate from the RGB image alone.

### E. Effect of Training-Set Size (H3)

To probe whether DS-YOLO is more data-efficient than the RGB baseline, we trained both variants on 25%, 50% and

TABLE V  
PER-COMPONENT-TYPE MAP@0.5 (BINARY OK/NG, RGB vs. DS-YOLO).

| Type                  | RGB      | DS-YOLO  | $\Delta$ (pp) |
|-----------------------|----------|----------|---------------|
| Chip resistor (R)     | [TODO: ] | [TODO: ] | [TODO: ]      |
| Ceramic capacitor (C) | [TODO: ] | [TODO: ] | [TODO: ]      |
| NE555 timer           | [TODO: ] | [TODO: ] | [TODO: ]      |
| CD4017 decade counter | [TODO: ] | [TODO: ] | [TODO: ]      |

100% of the training set and evaluated on the same test split. Fig. 2 shows that the gap between DS-YOLO and the RGB baseline grows as the training budget shrinks, supporting H3.

[data\_fraction.pdf – run step figures]

Fig. 2. mAP@0.5 on the test split as a function of training-set fraction. DS-YOLO consistently outperforms the RGB baseline; the advantage widens as the training budget shrinks, supporting H3.

#### F. Robustness to Alignment Errors

The benefit of **D** depends on the quality of the homography **H**. We synthetically perturbed **H** with random translations of standard deviation  $\sigma_t \in \{0, 2, 5, 10, 20\}$  px at inference time and measured the resulting mAP. We evaluate the RGB baseline (unaffected by alignment quality) and RG-YOLO (whose fourth-channel difference is directly degraded by misalignment). Results are reported in Table VI. RG-YOLO maintains its advantage over RGB up to  $\sigma_t \approx$  [TODO: 10] px and only drops below the baseline beyond  $\sigma_t \approx$  [TODO: 15] px—well above the residual registration error observed on the production cell ( $\sigma_t \leq$  [TODO: ] px in practice), confirming practical robustness.

#### G. Latency

End-to-end latency on the deployment PC ([TODO: CPU model, GPU model, RAM]) is reported in Table VII. We

TABLE VI  
ROBUSTNESS TO SYNTHETIC ALIGNMENT PERTURBATION (MAP@0.5).  $\sigma_t$  IS THE STANDARD DEVIATION OF THE RANDOM TRANSLATION ADDED ON TOP OF THE ORB HOMOGRAPHY AT INFERENCE TIME.

| $\sigma_t$ (px) | 0        | 2        | 5        | 10       | 20       |
|-----------------|----------|----------|----------|----------|----------|
| RGB baseline    | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| RG-YOLO         | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |

TABLE VII  
PER-STAGE LATENCY (MS), MEAN OVER 200 RUNS. [TODO: FILL FROM DEPLOYMENT PC MEASUREMENTS.]

| Stage                            | RGB      | RG-YOLO  | DS-YOLO  |
|----------------------------------|----------|----------|----------|
| Capture (camera + transfer)      | [TODO: ] | [TODO: ] | [TODO: ] |
| ORB + homography alignment       | [TODO: ] | [TODO: ] | [TODO: ] |
| Diff channel computation         | —        | [TODO: ] | —        |
| YOLOv8m forward (capture)        | [TODO: ] | [TODO: ] | [TODO: ] |
| YOLOv8m forward (golden, shared) | —        | —        | [TODO: ] |
| CRFM + neck + head               | —        | —        | [TODO: ] |
| Decision + OPC UA write          | [TODO: ] | [TODO: ] | [TODO: ] |
| Total                            | [TODO: ] | [TODO: ] | [TODO: ] |

compare the RGB baseline, RG-YOLO (one extra channel; difference computed on CPU) and DS-YOLO (second backbone pass on the golden image). The dominant cost is the YOLOv8m forward pass; the additional stages add [TODO: X] ms for RG-YOLO and [TODO: Y] ms for DS-YOLO relative to the RGB baseline.

#### H. Qualitative Results

Fig. 3 shows a representative golden image, the aligned capture and the resulting **D**. Defect locations stand out clearly in **D**, providing the detector with an explicit spatial prior. Fig. 4 compares the detections of the RGB baseline and DS-YOLO on a board containing one missing and one shifted component.

[qual\_diff.pdf – run step figures]

Fig. 3. From left to right: golden reference **I<sub>g</sub>**, the ORB-aligned capture  $\tilde{\mathbf{I}}$ , and the absolute-difference map **D** (visualised with a hot colormap). Defect regions light up clearly in **D** while remaining visually subtle in the RGB capture.

## ACKNOWLEDGMENT

**[TODO: Acknowledge funding source / lab equipment / etc.]**

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*[qual\_detect.pdf – run step figures]*

Fig. 4. Detection comparison on the same board: YOLOv8m RGB baseline (left) vs. DS-YOLO (right). Green boxes = OK components; red boxes = NG components. DS-YOLO correctly flags a shifted component (red) that the RGB baseline misses or assigns low confidence.

### I. Discussion and Limitations

The experiments support all three hypotheses, with the strongest gains on geometric defect classes and on small training budgets. Two limitations are worth noting. First, the method assumes that the golden reference is geometrically registrable to the capture; in practice this required a controlled fixture and a diffuse dome light, but no specific hardware beyond what is already present in standard AOI cells. Second, DS-YOLO inherits the classical limitation of supervised detectors: it can only flag defect modes that are represented in the training labels. Combining DS-YOLO with an unsupervised anomaly-detection branch (e.g. PaDiM **[TODO: cite]**) is a natural direction for future work.

## VII. CONCLUSION

We have presented Dual-Stream YOLO (DS-YOLO), a Siamese extension of YOLOv8m that exploits the golden reference image always available in an AVI cell by routing both the captured frame and the reference through a shared backbone and fusing their multi-scale features at  $P_3, P_4, P_5$  via a learnable Cross-Reference Fusion Module. The CRFM adds only  $\sim 70$  K parameters, preserves the rest of the backbone and is initialised so the network starts as the YOLOv8m baseline. On a real SMT production cell that integrates a Mitsubishi PLC, two articulated robots and an industrial GigE camera through CC-Link and OPC UA, DS-YOLO improves mAP@0.5 by +**[TODO:  $\Delta$ ]** pp over the RGB-only baseline, with the largest gains on geometric defect classes and small training budgets, and adds less than **[TODO:  $\delta t$ ]** ms to the per-board latency. Future work will investigate (i) end-to-end learning of the alignment together with the detection, (ii) integration of an unsupervised anomaly branch for unlabeled defect modes and (iii) extension to multi-view six-side inspection for full board coverage.